

#	Domain	AI FDE Challenge	Central Challenge	Key Enterprise Inputs	What the FDE Must Solve
				history, payment history, ERP logs	manipulation, approval bypass and suspicious transaction patterns while preserving segregation of duties, evidence and human escalation.

Sprint 2 - Deliverables with Presentation by teams

#	Expectation	Expected Outcome
1	Use Repo 1.0 as the As-Is baseline	Understand the existing case-study repository, current architecture, data, workflows, gaps, inconsistencies, risks and technical debt.
2	Transform Repo 1.0 into Repo 2.0 (Use Prompt Engineering + Context Engineering with any LLM of your choice)	Solve the problems inherent in Repo 1.0 by improving the architecture, AI workflow, risk controls, compliance, governance, security, observability, resilience and production readiness.
3	Convert Repo 2.0 into a PRD	Produce a complete Product Requirements Document covering business problem, users, functional requirements, AI capabilities, workflows, data, integrations, risk/compliance requirements, non-functional requirements, acceptance criteria and success metrics.
4	Develop a Full-Fledged Working Application	Build and demonstrate a functional application based on the PRD using Google AI Studio Build or any tool/platform of choice, with working workflows, AI capabilities, decision logic,

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		controls, explainability and an end-to-end demo.

These are intentionally problematic As-Is environments.

They contain contradictions, incomplete evidence, data-quality issues, weak model evaluation, unsafe fallbacks, human-oversight gaps, compliance/audit weaknesses, security issues, observability gaps, and production-readiness debt.

They do not contain Repo 2.0 solutions (That's your job as AI FDE)

1. Manufacturing / Automotive — Supplier Quality & Incoming Material Release
2. Insurance / BFSI — Claims Fraud, Triage & Investigation
3. Energy & Utilities — Field Work Safety & Maintenance Authorization
4. Life Sciences / Pharma — Batch Deviation Investigation & Disposition
5. Enterprise ERP / Shared Services — Procure-to-Pay Exception & Payment Control

Each repository follows the same structural spine:

Decision & Risk Boundary → Data & Evidence → Detection Models → Document/Identity/ Compliance → Workflow Orchestration → Governed Decisioning → Explainability & Review → Compliance & Audit → AI Risk/Security/Observability → Production Readiness

This should make them particularly suitable for the workshop flow:

Repo 1.0 → Prompt Engineering + Context Engineering → Repo 2.0 → PRD → Working AI FDE Application.